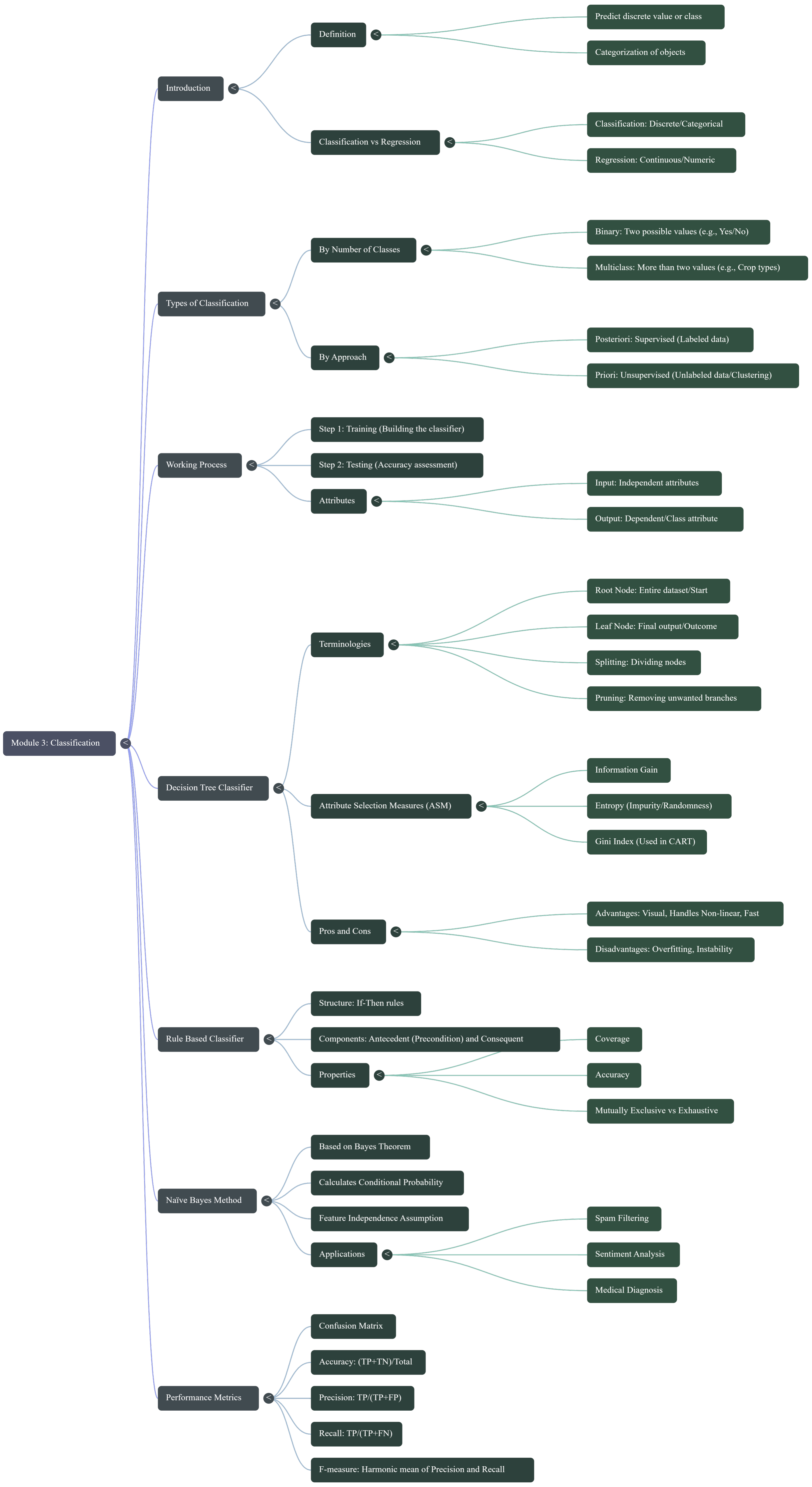

Module 3: Classification
Introduction
Definition
Predict discrete value or class
Categorization of objects
Classification vs Regression
Classification: Discrete/Categorical
Regression: Continuous/Numeric
Types of Classification
By Number of Classes
Binary: Two possible values (e.g., Yes/No)
Multiclass: More than two values (e.g., Crop types)
By Approach
Posteriori: Supervised (Labeled data)
Priori: Unsupervised (Unlabeled data/Clustering)
Working Process
Step 1: Training (Building the classifier)
Step 2: Testing (Accuracy assessment)
Attributes
Input: Independent attributes
Output: Dependent/Class attribute
Decision Tree Classifier
Terminologies
Root Node: Entire dataset/Start
Leaf Node: Final output/Outcome
Splitting: Dividing nodes
Pruning: Removing unwanted branches
Attribute Selection Measures (ASM)
Information Gain
Entropy (Impurity/Randomness)
Gini Index (Used in CART)
Pros and Cons
Advantages: Visual, Handles Non-linear, Fast
Disadvantages: Overfitting, Instability
Rule Based Classifier
Structure: If-Then rules
Components: Antecedent (Precondition) and Consequent
Properties
Coverage
Accuracy
Mutually Exclusive vs Exhaustive
Naïve Bayes Method
Based on Bayes Theorem
Calculates Conditional Probability
Feature Independence Assumption
Applications
Spam Filtering
Sentiment Analysis
Medical Diagnosis
Performance Metrics
Confusion Matrix
Accuracy: (TP+TN)/Total
Precision: TP/(TP+FP)
Recall: TP/(TP+FN)
F-measure: Harmonic mean of Precision and Recall